

Household Energy Poverty and Time Allocation: Empirical Evidence from China

Abstract: Sufficient leisure time and appropriate working time contribute to individual well-being and income respectively, therefore clarifying the relationship between energy poverty and individual time allocation has a profound impact on socio-economic development. This study explores the relationship between energy poverty and individual time allocation using China Family Panel Studies (CFPS) data. We found that household energy poverty significantly reduces individual working time and leisure time, but increases the time for housework. Life attitudes, health status, employment, and single status could be potential mechanisms to explain how household energy poverty affects individual time allocation. Our results demonstrate that household energy poverty reduces the effective time allocation of individuals, resulting in a waste of human resources, which may hinder social sustainable development. We proffered policy recommendations based on these findings in the end, which provide an empirical reference for developing countries worldwide.

Keywords: Time Allocation; Household Energy Poverty; Working Time; Housework Time; Leisure Time

1. Introduction

The accessibility and affordability of modern energy carry substantial implications for human health and economic growth (Zhang et al., 2019; Koomson and Danquah, 2021; Wang, 2023). However, numerous households worldwide still lack access or ability to use sufficient, secure, clean, and environmentally friendly modern energy—a circumstance generally referred to as household energy poverty (Charlier, 2019; Dogan et al., 2022; Luan et al., 2023). Household energy poverty poses a negative threat to household members' education, job performance, and health (Sharma et al., 2019; Zhang et al., 2019). For example, due to energy poverty, particularly the absence of electricity, children face difficulties in studying during the evening hours, this limitation could additionally impact their academic achievements (Zhang et al., 2019). Additionally, the lack of access to modern cooking energy such as electricity or natural gas may cause indoor air pollution, which may contribute to lung cancer, chronic obstructive pulmonary disease, and other diseases (Smith et al., 2004; Bukari et al., 2021). Consequently, energy poverty influences education, employment, and health, which may shape individuals' time allocation. This paper investigates how household energy poverty affects the way individuals allocate their time and explore the underlying channels.

Why the study of energy poverty and its impact on individual time allocation is compelling? That comes from efficient time allocation. In other words, eradicating energy poverty may decrease the time spent on housework, and further increase leisure time and working time. For example, alleviating accessibility to energy poverty could reduce the time spent on cooking and collecting conventional fuel. Therefore, efficient time allocation may have twofold benefits. For one thing, it can liberate some time and then redistribute to leisure or working time which can bring well-being and create economic value. For another, it also contributes to reducing intra-household conflicts (Devoto et al., 2012). From the perspective of household energy use types, for households that use solid fuel, it is common to spend a lot of time collecting firewood or transporting coal, and then burning it for cooking and heating, which makes housework less efficient and crowds out leisure

time. For households that use modern energy at high cost, the efficiency of kitchen appliances may be low (McNeil et al., 2008; IEA, 2022), which may lead to more housework time. Therefore, household energy poverty affects the efficiency of housework activities, leading to a potential increase in housework time, which may further reduce individual working time and leisure time. In contrast, reducing working and leisure time harms personal well-being and social value creation. Therefore, an in-depth analysis of the influence of energy poverty on individual time allocation is an issue that deserves study.

Energy poverty is a worldwide issue, but the specific definition and measure of energy poverty have not yet reached a consensus (Moore, 2012; Faiella and Lavecchia, 2021). In addition, energy poverty performs a little bit differently across developed nations and developing countries. Energy poverty in developing countries is mainly from the perspective of accessibility of modern energy sources as some households lack the opportunity to access adequate energy services, whose evidence includes Zhang et al. (2019) in China, Ntaintasis et al. (2019) in Greece, Abbas et al. (2020) in six South Asian countries. Meanwhile, energy poverty in developed countries is mainly from the viewpoint of affordability of modern energy sources because the energy services are common but high cost for some households, whose evidence comes from Castano-Rosa et al. (2019) in European Union, Betto et al. (2020) in Italy, Karpinska and Smiech (2020) in Poland, Churchill and Smyth (2021) in Australia, Dogan et al. (2022) in America. Nowadays, due to the pandemic and the energy crisis, approximately 75 million individuals have lost access to affordable electricity services, and around 100 million people have been unable to afford clean cooking solutions around the world¹ (IEA, 2022). Additionally, the type of household energy, especially the type of cooking energy, will largely determine the efficiency of household activities, the use of solid energy such as firewood or coal may reduce housework efficiency, while modern energy such as electricity, solar energy, natural gas may improve housework efficiency. Therefore, the type of energy used may potentially affect individual time allocation.

At present, the energy poverty literatures are approximately divided into two categories, one is to explore the sources of energy poverty, and the other is to study the impacts caused by energy poverty. The former study mainly discusses the influence of household wealth, property ownership, socio-economic status, geographical environment, race, retirement income, and other factors on energy poverty (Marchand et al., 2019; Abbas et al., 2020; Zou et al., 2020; Dogan et al., 2022; Fry, 2022). The later study mainly investigates the impact of energy poverty on health, public health, income, education, subjective well-being, job performance, and other factors (Bridge et al., 2016; Oum, 2019; Sharma et al., 2019; Zhang et al., 2019; Churchill et al., 2020). However, the impact of household energy poverty on an individual time allocation has yet to be studied, possibly due to a lack of appropriate data and measures. This study fills this gap by using the China Family Panel Studies (CFPS) database collected by Peking University and the previous energy poverty measures to explore the relationship between energy poverty and individual time allocation. Energy poverty is caused by a combination of factors ranging from the lack of availability to the insufficient affordability of modern energy. About 52.46% of rural households still use solid fuels (firewood or coal) for cooking (Zhang et al., 2019), even if the electricity coverage is 100% in China. Additionally, many urban households are already dependent on modern energy, but lower-income groups tend to use older, less efficient appliances in China, which leads to affordability issues (IEA, 2022). Therefore, we adopted the multidimensional measures according to Luan et al. (2023) to define

¹ Source: <https://www.iea.org/reports/world-energy-outlook-2022>

household energy poverty.

Consequently, we attempt to investigate the following questions. First, how does household energy poverty affect individual time allocation (working time, housework time, and leisure time)? Second, is there a group heterogeneity in the effect of energy poverty on individual time allocation? Third, what are the possible channels of energy poverty affecting individual time allocation? Accordingly, some findings of this study are listed as follows.

1. Household energy poverty significantly reduces an individual working time and leisure time, but increases housework time, the results are robust against instrumental variables, different energy poverty measures, and an alternative regression model.

2. The impact of household energy poverty on individual time allocation is heterogeneous. The heterogeneous findings indicate that (1) household energy poverty has a greater negative impact on reducing working time in the male group, individuals aged below 45 years old, people who have Agricultural Hkou; (2) household energy poverty has a greater positive influence on increasing housework time in the male group, individual aged below 45 years old; (3) the household energy poverty has a greater negative impact on reducing leisure time in individual aged below 45 years old.

3. We find four potential channels (including life attitudes, health status, employment, and single status) that may through household energy poverty influence individual time allocation. These mechanism analyses provide a better understanding of how household energy poverty influences individual time allocation.

This study contributes to the literature in the following aspects. First, to the best of our knowledge, no study in the literature has empirically examined the relationship between household energy poverty and individual time allocation, thus this study fills this gap. Second, especially from the perspective of sustainable development in economics (Basch, 2014), this paper finds that household energy poverty greatly increases individual time spent on housework, which undoubtedly reduces the effective use of individual time. Based on previous studies, this paper gives empirical evidence that household energy poverty hinders sustainable development in economics from the perspective of time allocation, which can also provide a reference for the efficient use of individual time in other developing countries. Third, we use the elevation drop as an instrumental variable of energy poverty accessibility, which can better alleviate the endogenous problem and improve the credibility of the baseline conclusions, which is an empirical improvement over the existing energy poverty literature. Finally, this paper carries out four possible mechanisms and finds a lot of interesting conclusions. All four of these factors may be channels of energy poverty that reduce the effective use of individual time, which is consistent with the findings of energy poverty research in China (Zhang et al., 2019; Nie et al., 2021; Zhang et al., 2021). This also complements the existing relevant research literature.

The rest of this paper is organized as follows. Section 2 briefly reviews the literature on the measure of energy poverty, the source of energy poverty, the influence of energy poverty, and time allocation. Section 3 details the data, variables, and methodology used in the analysis. Section 4 presents the baseline empirical results and robustness checks. Section 5 conducts the heterogeneity analysis. Section 6 discusses the possible channels. Section 7 concludes this study and states policy implications.

2. Literature review

To better explore the relationship between energy poverty and individual time allocation, we

review the following related literatures: the measure of energy poverty, the impact of energy poverty, the influence factors of individual time allocation.

2.1 The measure of energy poverty

In China, energy consumption in households is generally used for lighting, cooking, temperature maintenance, and other household appliances. Although the coverage rate of electricity is 100% in China, a proportion of households suffer energy poverty. The words of “energy poverty” has been proposed for a long time, but neither the definition nor the measurement of energy poverty has reached consensus (Lewis, 1982; Zhang et al., 2019). Lewis (1982) firstly introduced the notion of energy poverty to illustrate how a lack of adequate energy utilization can affect individuals living standards. Since then, many scholars have offered their insights into energy poverty, but these definitions can be approximately divided into two categories: accessibility and affordability. The measurement of energy poverty is also consistent with the definition. Following the logic of accessibility, the household is classified as suffering from energy poverty if the main cooking fuel is firewood or coal (Adusah-Poku and Takeuchi, 2019; Zhang et al., 2019; Luan et al., 2023). Following the logic of affordability, the household is classified as suffering from energy poverty if ratio of the expenditure to the household income is greater than 10% (Boardman, 1991; Barnes et al., 2011; Bouzarovski and Petrova, 2015; Churchill and Smyth, 2020), while some others used the alternative thresholds, such as 5%, 15%, and 20% (Churchill and Smyth, 2020). To avoid high-income household with overconsumption in energy mistakenly classified as suffering energy poverty, scholars first identify low-income household, which the income is below the median among the district, then determine the true energy poverty through expenditure threshold (Hills, 2012; Lin and Wang, 2020). There is a common sense that the accessibility is more suitable for developing countries, while the affordability is more resealable for developed countries.

However, recent research has begun to explore multidimensional measurement. For instance, Nussbaumer et al. (2013) introduced the Multidimensional Energy Poverty Index (MEPI) and explored this matter within the context of developing economies. Similarly, Sadath and Acharya (2017) proposed a straightforward multidimensional gauge of energy poverty specific to India. These investigations primarily address accessibility issues and encompass various energy sources in their multidimensional assessments. Comparable multidimensional approaches to understanding energy poverty can be observed in Sher et al. (2014) and Ogwumike and Ozughalu (2016). Some others combine affordability with accessibility to measure energy poverty (Zhang et al., 2019; Luan et al., 2023).

In addition, recent research has begun to use micro level data to measure energy poverty. For macro level, Tang and Liao (2014) capture the energy accessibility poverty based on national population census data and found that there were approximately 490 million rural households and 170 million urban households suffering energy accessibility poverty in China in 2010. Wang et al. (2015) propose a comprehensive index to measure the energy poverty in China, which composes energy service availability, energy consumption cleanliness, energy management completeness, and household energy affordability and energy efficiency. Pan et al. (2021) take the World Development Indicators (WDI) to measure the energy poverty for numerous countries around the world. For micro level, Ampofo and Mabefam (2021) takes individual-level data from World Values Survey (WVS) to measure energy poverty in numerous countries. Additionally, household-level survey data are also widely used to capture energy poverty. Such as Lin and Wang (2020) use the electricity bills and usage of electricity from Chinese Residential Energy Consumption Survey (CRECS) to measure

household energy poverty; Rafi et al. (2021) adopt household-level India Human Development Survey (IHDS) data to estimate the multidimensional energy poverty in India. Similarly, the China Family Panel Studies (CFPS) has also been widely used in research on energy poverty in China (Zhang et al., 2019; Nie et al., 2021; Zhang et al., 2021; Luan et al., 2023). Koomson and Danquah (2021) employ the weighted method to describe the multidimensional energy poverty based on the micro-level Ghana Living Standard Survey. In summary, recent literatures have begun to use micro-level data, particularly at the household-level, to measure household energy poverty.

2.2 The impact of energy poverty

WHO (2007) reported that indoor pollution from the use of stationary fuels is a significant contributor to disease and death in poor countries. Therefore, energy poverty can have a wide and important impact on the economic, cultural, and population quality development of the whole society (Green and Gilberston, 2008; Acharya and Sadath, 2019). At present, there are many papers that investigate the impact of energy poverty on health, public health, income, education, subjective well-being, job performance, and other factors (Bridge et al., 2016; Oum, 2019; Sharma et al., 2019; Zhang et al., 2019; Churchill et al., 2020). Zhang et al. (2021) proposed a multidimensional energy poverty indicator to explore the relationship between energy poverty and health, then found the energy poverty can significantly decrease health. Similarly, Churchill and Smyth (2021) found energy poverty is one of the casual factors for adverse health due to the households suffering energy poverty and unable to maintain adequate warmth in the winter. At the same time, energy poverty can also have an indirect impact on mental health, leading to a greater probability of consequences such as depression, poor spirituality, and negative life perceptions (Druica et al., 2019; Nie et al., 2021). Furthermore, Pan et al. (2021) proposed the negative impacts of energy poverty on public health, not only reducing life expectancy but also increasing the probability of death based on the World Development Indicators (WDI) database.

2.3 The influence factors of individual time allocation

For the study of factors affecting individual time allocation. First, from a macro perspective, the overall level of economic development of society, the social employment situation, government employment policy control, the impact of health emergencies, and traditional cultural ideas (e.g., Confucianism) are all potential factors that affect individual time allocation (Hersch and Stratton, 2002; Zhang Jinhua and Hu Junhui, 2012; Chen and Li, 2014). Second, from a micro perspective, individual characteristics and family characteristics have a significant impact on individual time allocation. On the one hand, individual characteristics are the primary factors for individuals to weigh their time allocation, among which gender, age, education level, non-cognitive ability, and nature of work are the basic determinants of whether an individual is able to choose a desirable job, whether he or she is able to freely organize his or her work, and how much leisure time he or she has (Becker, 1960; 1976; Hersch and Stratton, 2002). On the other hand, the family, as the basic social unit in which individuals live, work, and have children, its family size, number of children, gender of the children, whether it supports the elderly, whether there is intergenerational economic help, and intergenerational rearing and caring are all factors that influence couples in making the work-housework-leisure trade-off (Mankiw, 2014 ; Guo et al, 2018; Sun, 2021; Guo, Kaiming et al, 2021).

The motivation of this study. Based on these studies above, we can find that: from the perspective of the family unit, although energy poverty at the micro-family level is also one of the

important characteristics of the family, there is little literature that does a study of the impact of family energy poverty on the time of the individual. The allocation of household time in the society is closely related to the high-quality development of the population worldwide, because wasting time on unpaid housework and collecting solid fuel undoubtedly leads to a waste of human resources. On the one hand, the increased time spent on housework will decrease working hours, which may reduce family and individual income; on the other hand, this may also reduce individual leisure time, which in turn reduces individual subjective well-being (Zhang et al., 2023). Nowadays, society attaches great importance to the well-being of people, and there is a lack of exploratory work in this area for the existing research in China, which undoubtedly inspires the research motivation of this paper.

3. Data and variables

3.1 Introduction to CFPS

The data used in this study are from China Family Panel Studies (CFPS), a nationally micro survey started by Peking University in 2010. The survey collects data at the community, household, and individual levels from 25 provinces in mainland China, to support the academic research on China's society, economy, population, and education. Meanwhile, CFPS has also been widely used in research on household energy poverty in China (Zhang et al., 2019; Nie et al., 2021; Zhang et al., 2021). It currently includes six waves: 2010, 2012, 2014, 2016, 2018, 2020. In order to exclude the impact of public health emergencies (COVID-19) and maintain the continuity of research data², we use the survey data from 2014, 2016 and 2018 for research.

3.2 Construction variables

3.2.1 Household energy poverty

Household energy poverty is the key independent variables in this study, it had multiple measures in previous studies. However, this study applies a combination of multidimensional household energy poverty indicators of accessibility and affordability, because China has both accessibility and affordability issues. For the accessibility, although the electricity is fully covered in China, a large number of households still cannot access modern cooking energy. The frequency and percentage of different types of cooking fuel used in Chinese households is detailed in Fig. 1. There are 4921, 4387, 3675 households (Fig. 1), approximately accounting 46.10%, 40.30%, 33.55% that still using firewood or coal for cooking respectively across three years in the sample (Fig. 2). For the affordability, energy consumption in households is generally used for lighting, cooking, temperature maintenance, and other household appliances. The electricity and fuel are the main energy forms to support these daily needs, but the expenditure on it may be a heavy burden for some households. The ratio of energy expenditure (electricity plus fuel) to household income over 10% is 21.62% in 2014 but falls to 21.18% in 2018³ (Fig. 3). Therefore, the combination of accessibility and affordability is suitable for measuring household energy poverty in China.

² Data for 2010 and 2012 lacked separate information on household electricity and fuel expenditures, so only data for 2014, 2016 and 2018 were used.

³ The energy expenditure includes electricity and fuel, we replaced the ratio of energy expenditure greater than 1 with 1 to avoid the influence of outliers. These data are different from Zhang et al. (2019) as we only include electricity and fuel.

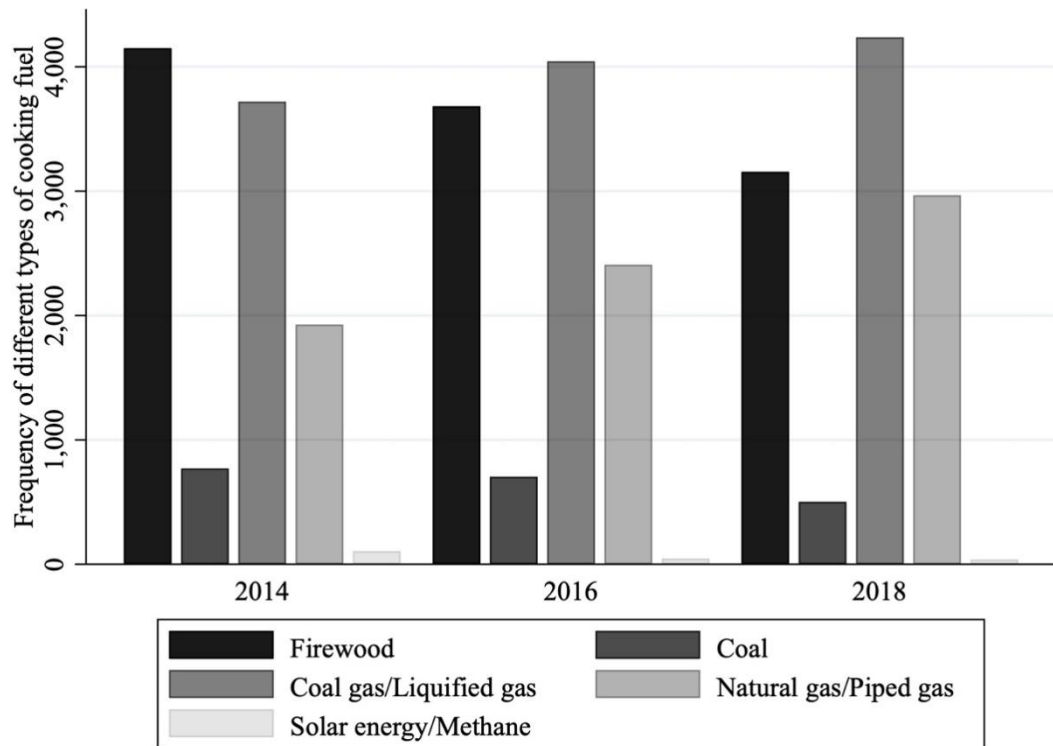


Fig. 1. The frequency of different types of fuel used for cooking in China (2014-2018).

Source: China Family Panel Studies.

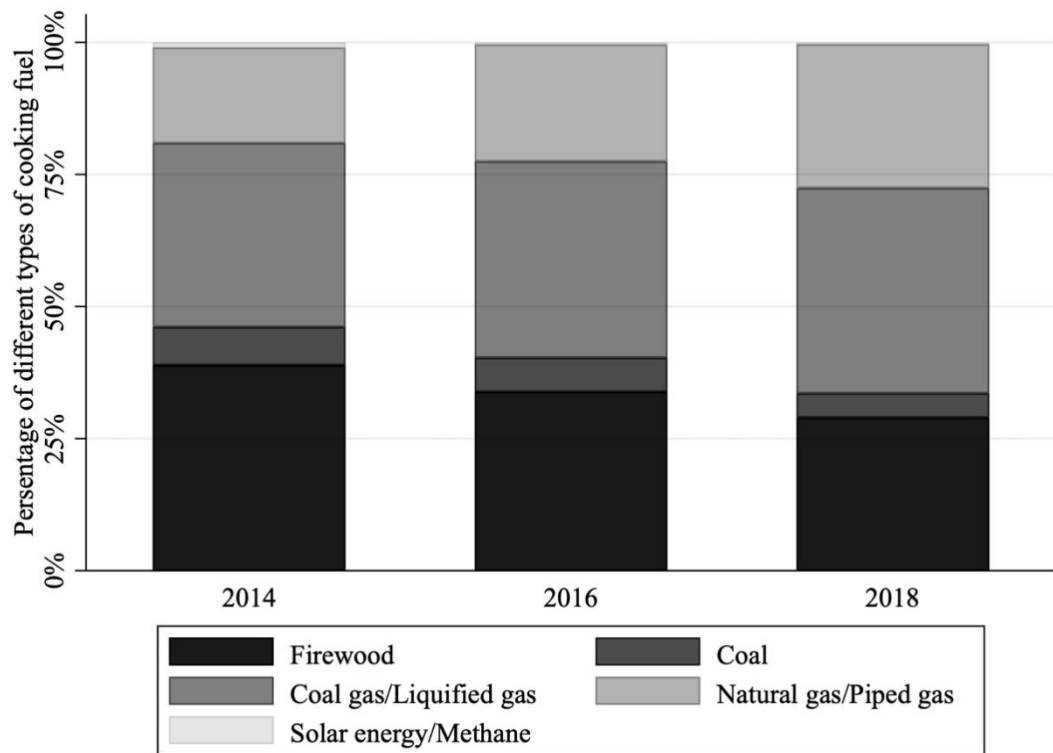


Fig. 2. The percentage of different types of fuel used for cooking in China (2014-2018).

Source: China Family Panel Studies.

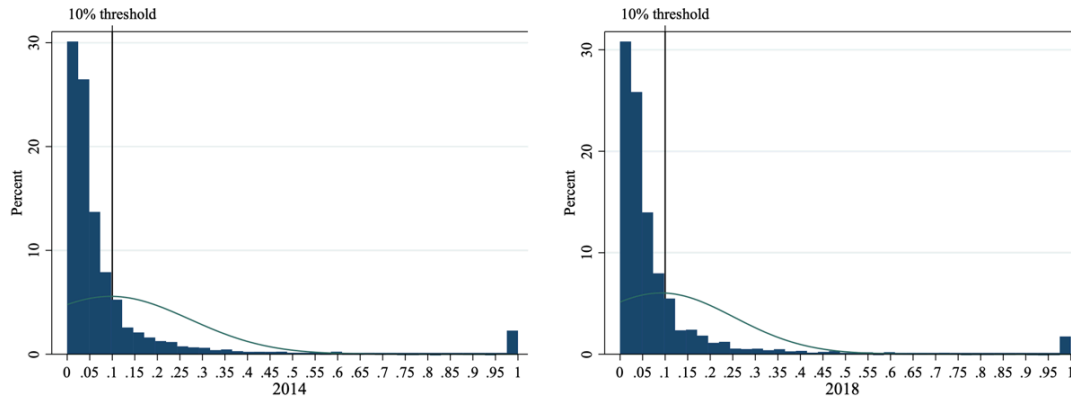


Fig. 3. The ratio of energy expenditure to household income (2014, 2018).

Source: China Family Panel Studies.

Referring to Luan et al. (2023), this study measures accessibility based on the question “What is the main source of cooking fuel in your household?” in the questionnaire. If the main cooking fuel is solid fuel (firewood or coal), the household is classified as having energy accessibility poverty, Accessibility=1; otherwise, Accessibility=0. Additionally, for the energy affordability poverty, we adopt the low income-high cost (LIHC) framework proposed by Hills (2012) and recommended by Lin and Wang (2020), to avoid misclassifying high-income households with energy overconsumption as part of the energy affordability poverty group. If the household income is lower than the median household income for the county, the household is considered low-income. Furthermore, combining 10% threshold, if the ratio of energy expenditure to household income is higher than 10% and the household is low-income, thus the household is classified as having energy affordability poverty, Affordability=1; otherwise, Affordability=0. Consequently, we propose the Family Multidimensional Energy Poverty Indicator (FMEPI) based on Accessibility and Affordability. When both Accessibility and Affordability are 0, FMEPI=1; when one of Accessibility and Affordability is 0 and the other is 1, FMEPI=2; when both Accessibility and Affordability are 1, FMEPI=3. FMEPI=1 indicates that there is no energy poverty; FMEPI=2 indicates that there is one type of energy poverty; FMEPI=3 indicates that there are two types of energy poverty. The specific definitions of FMEPI reported in Table 1.

3.2.2 Individual time allocation

Individual time allocation is the key dependent variables in this study, the working time, housework time, and leisure time come from adult questionnaire. The working time was obtained from the question “In the past 12 months, on average, how many hours per week have you worked in your current job (excluding lunch breaks but including overtime)?” (in hours). For individuals who reported no employment, their work hours were recorded as zero. The housework time was obtained based on the question “Generally, how many hours do you spend on housework (excluding caregiving)?” To facilitate subsequent comparisons and analyses, all time measures were converted into weekly. The measure of leisure time is not consensus. Some scholars argue that leisure time primarily refers to the time individuals spend engaging in leisure activities such as socializing, physical exercise, gaming, cultural experiences, and similar activities (Kelly and Godbey, 1992; Kleiber et al., 2011). Others argue that leisure time mainly encompasses the time individuals perceive as leisure activities based on their own judgment (Kleiber et al., 2011; Russell, 1987; Tinsley and Eldredge, 1995). The key distinction between the two perspectives is that the latter is

subject to individual perceptions and experiences, potentially resulting in variations in how leisure activities are defined. To ensure consistency in measuring leisure time and avoid discrepancies arising from individual perceptions or experiences, this study adopts the former approach. It does not include individual sleep and other minor fragments of time. Therefore, following the approach proposed by Zhang et al. (2023), individual weekly leisure time was obtained from the sum of these questions: “How many hours do you spend watching television or movies per week?”, “How many hours do you spend exercising per week?”, and “How many hours do you spend on the internet per week?”.

Fig. 4 presents the intuitive relationship between Family Multidimensional Energy Poverty Indicator (FMEPI) and individual time allocation (Working Time, Housework Time, and Leisure Time). Here, working time and leisure time increase as FMEPI increases and leisure time decreases as FMEPI increases. However, the relationship here may be influenced by other variables, so we further explore the clean relationship between FMEPI and individual time allocation by adding control variables and controlling fixed effects.

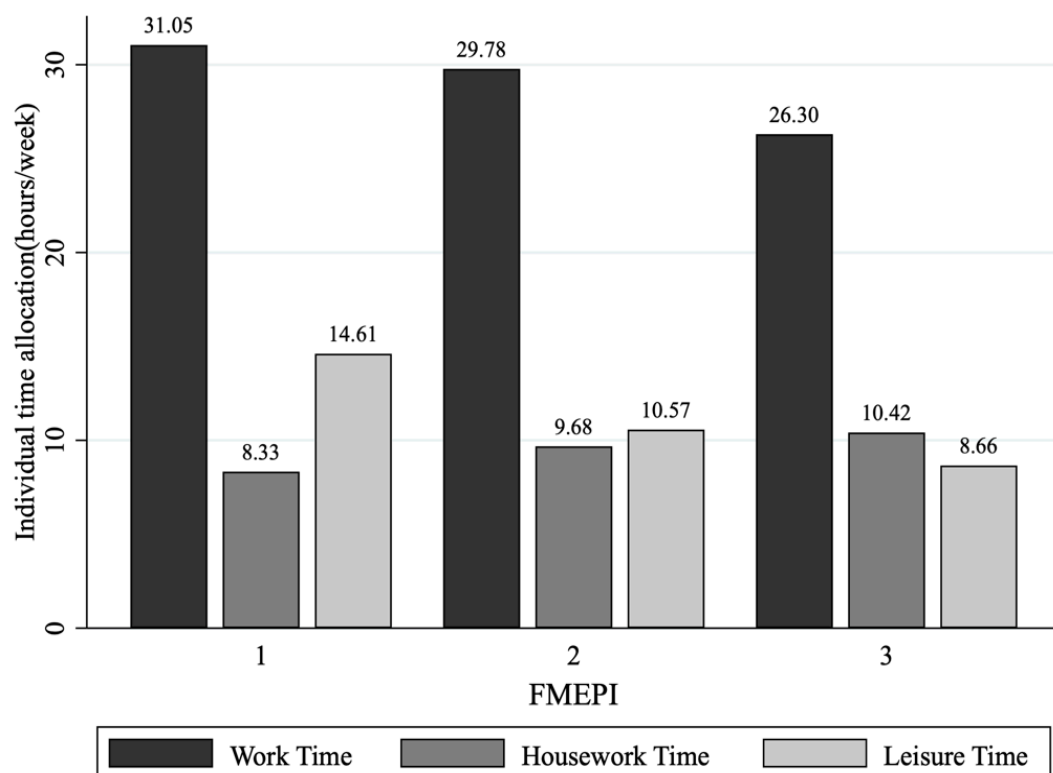


Fig. 4. The relationship between FMEPI and individual time allocation (2014-2018)

Source: China Family Panel Studies.

3.2.3 Control variables

There are several factors that may influence individual time allocation, and in order to alleviate biased estimation of the baseline regression analyses due to omitted variables, this paper controls for observable variables. Referring to Luan and Huang (2023), the control variables include the individual characteristics of the respondent (gender, age, education level, household type, health level, and personal income level) and household characteristic (household members, household per capita annual income, and the ownership of the house). The number of family members comes from

the Family Relationship Pool, and the per capita household income and the ownership of the house come from the Family Economy Pool. In terms of sample selection, given the potential substitutability between working time, housework time, and leisure time, this study primarily focused on the working-age population. Notably, a significant segment of the individuals remains actively engaged in employment until they are 70⁴. Within the three-year China Family Panel Studies (CFPS) dataset employed for this investigation, individuals aged between 60 and 70 constituted a notable proportion, exceeding 13% of the whole sample. Additionally, compulsory education is only nine years in China, so many students who unable to access high school or dropout from technical schools start working at the age of 16. Therefore, this study selected individuals aged between 16 and 70.

3.2.4 Descriptive statistics analysis

Table 1 presents descriptive statistics of variables. Regarding individual working time, the average weekly working hours are approximately 25 hours, which is lower than the figures reported by Zhang et al. (2023). This discrepancy can primarily be attributed to the inclusion of individuals without employment in our sample. Weekly housework time are approximately 9 hours, while weekly leisure time does not exceed 13 hours, which indicates Chinese residents spend slightly over an hour on housework but have less than two hours for leisure each day. Regarding the Family Multidimensional Energy Poverty Indicator (FMEPI), the value is approximately 1.5. This statistical value aligns closely with previous literature (Zhang et al., 2019; Luan et al., 2023). In the full sample, the gender distribution is roughly balanced, with an average age around 41 years old. The household size ranges of 4 to 5 people, consistent with existing research findings.

Table 1 Summary Statistics

Variables	Description	Mean	Std. Dev	Obs
Independent variables				
Working Time	Working hours per week	25.03	27.73	60111
Housework Time	Housework time per week	9.13	12.09	60111
Leisure Time	Leisure time per week	12.61	15.26	60111
Dependent variable				
FMEPI	Multidimensional measure of household energy poverty. FMEPI =1, no firewood or coal for cooking, ratio of energy consumption to total income less than 10%, household income greater than median income FMEPI =3, cooking with firewood or coal, ratio of energy consumption to total income greater than 10%, household income less than median income FMEPI =2, otherwise	1.54	0.65	60111
Individual characteristics				
Gender	Male=1, Female=0	0.51	0.50	60110
Age	Age (in years)	40.82	14.64	60111
Hukou	Type of registered permanent residence. Agriculture=1, Non-agriculture =2.	1.26	0.44	60111
Education	Years of education	2.98	1.62	59985
Status	Social status in the local area (very low 1-2-3-4-5 very high)	2.90	1.00	38657

⁴ In China, women usually retire at age 55 and men at age 60.

Variables	Description	Mean	Std. Dev	Obs
Household characteristics				
Family Size	Number of family members	4.46	1.95	60111
F_income	Average annual household income per capita (in RMB 10,000)	1.43	1.43	60111
Proportion16	Proportion of children aged 16 and below in the household	2.55	1.45	60111
Proportion65	Proportion of the elderly aged 65 and above in the household	0.11	0.22	60111

Note: The data presented above consists of longitudinal cross-sectional data from the China Family Panel Studies (CFPS) for the years 2014, 2016, and 2018.

3.2 Identification strategy and endogeneity

The estimation of the relationship between household energy poverty and individual time allocation requires an identification strategy. The identification model is specified as an OLS regression of the form:

$$Time_{cit} = \alpha + \beta FMEPI_{cit} + \gamma X_{cit} + \mu_{ih} + \tau_t + \varepsilon_{cit} \quad (1)$$

where the subscript $i = 1, 2, \dots, N$ represents individual; $c = 1, 2, \dots, N$ refers to the village or community the individual live; $t = 2014, 2016, 2018$ indicates the time of data collection. $Time_{cit}$ stands the individual i 's time allocation who lived in village/community c in year t , including weekly working time, weekly housework time, and weekly leisure time. $FMEPI_{cit}$ denotes the family multidimensional household energy poverty index of individual i who lived in village/community c in year t . X_{cit} is a vector of control variables including individual characteristics (Gender, Age, Hukou, Education, status) and household characteristics (Family Size, F_income, Proportion16, Proportion65). μ_{ih} signifies the county fixed effect, τ_t refers to the year fixed effect, ε_{cit} denotes the idiosyncratic error term. The coefficient of interest, β , captures the response of individual working time, housework time, and leisure time to household energy poverty.

However, this study may face endogeneity problems that come from two aspects. On the one hand, there may exist a causality problem. In other words, individual time allocation may affect their household energy poverty situation. For instance, if a person spends more time on work that increases income or on energy-saving measures, it could alleviate energy poverty in their household. In this case, there is a causal relationship between individual time allocation and household energy poverty, leading to endogeneity issues. On the other hand, unobservable omitted variables such as traditional social norms, such as thrifty, cause biased regression estimation. Historically, the cultures of Confucianism and Taoism both emphasize thriftiness, developing a habit of saving among the Chinese population from ancient times. Additionally, the competitive saving motive also contributes to a notably high savings rate among Chinese residents (Wei & Zhang, 2011). Consequently, even households that can access and afford clean energy might opt to continue using free firewood or low-cost coal for reasons of cost-saving. Thus, self-selection results affect individual time allocation.

We address these potential concerns with an instrumental variable approach. First, the average energy poverty share of the county (AEPS) is adapted as an instrumental variable for household energy poverty, which follows the logic of Zhang et al.(2019). Specifically, the correlation comes from the high pertinence between regional energy poverty and household energy poverty (Okushima, 2019). In terms of exogeneity, the AEPS is independent of unobserved variables other than the explanatory variables. Moreover, AEPS does not exert a direct effect on individual time allocation other than its effect through household energy poverty. In this case, AEPS is a county-level indicator that can not pose an influence on individual behavior, but it is highly correlated with household

energy poverty that can directly affect individual behavior. Therefore, AEPS is a valid instrumental variable.

Second, we also utilize the elevation drop as an instrumental variable for household energy poverty. For the correlation, it is intuitively obvious that energy poverty is positively related to elevation drop. At high altitudes, the complex terrain and fragile ecological environment make energy infrastructure construction difficult. Not only is it more difficult to construct, but it is also more difficult for construction personnel and materials to reach the construction sites, resulting in a significant increase in costs. In addition, complex terrain and fragile ecosystems also make maintenance difficult and costly. Therefore, due to high construction and maintenance costs, there are fewer energy construction projects at high altitude areas in China. As a result, people living in high altitudes areas have less access to modern energy or use them at higher costs, which can lead to household energy poverty. For the exogeneity, the elevation drop of residence is a natural geographical feature, which is completely unaffected by individual behavior and satisfies the hypothesis of exogeneity. In terms of exclusivity, the existence of high mountains and hills makes the elevation drop of some counties particularly large, but Chinese residents live and work in relatively flat areas mostly, and the elevation drop between their living and working locations is not much different, so the concern that the local elevation drop directly affects the personal time allocation is unlikely to exist. Therefore, we adopted these two instrumental variables to address the endogeneity issue.

4. Empirical results

This section is structured as follows. Firstly, this study estimates the impact effect of FMEPI on individual time allocation using the Ordinary Least Squares (OLS) method. Secondly, this study employs the Instrumental Variable (IV) approach to estimate the causal impact effect of FMEPI on individual time allocation. Finally, alternative variable and alternative model methods are utilized to conduct robustness checks.

4.1 FMEPI and individual time allocation

We begin our analysis with OLS estimation. Although the OLS suggestive the correlation between FMEPI and individual time allocation, they may inflate the true effect of FMEPI due to the presence of omitted variables and not taking consideration of time-invariant variables with time-invariant effects. To solve these issues, we include county and year fixed effects. Table 2 presents the baseline results clustered at the county level for robust standard errors, and each column controls for county fixed effects. Specifically, columns (1), (3), and (5) control for individual characteristics, year fixed effects, and county fixed effects. Columns (2), (4), and (6) additionally include household characteristics. Table 2 reveals that the coefficients of FMEPI on the three dimensions of individual time allocation are all statistically significant at 1% level. From the perspective of working time, as FMEPI increases by 1%, the individual weekly working time decreases significantly by 0.553 hours. In terms of housework time, with a 1% increase in FMEPI, the individual weekly housework time increases significantly by 0.946 hours. Regarding leisure time, a 1% increase in FMEPI leads to a reduction of 0.820 hours in an individual weekly leisure time. Regarding control variables, males tend to allocate more time to work and leisure but devote less time to housework, aligning with the traditional “men as breadwinners, women as homemakers” norm in China. Additionally, the age factor remains consistent with previous research, revealing a decrease in both working time and leisure time with age, alongside an increase in housework time (Zhang et al., 2023). Higher levels

of education tend to spend more time on leisure and less time on housework. Non-rural household registration is associated with less working time and housework time but more leisure time, while family income is positively correlated with housework time and leisure time. As the number of household members increases, housework time also increases.

Table 2

Energy Poverty and Individual Working Time, Housework Time, and Leisure Time (Baseline OLS regression).

Variables	Working Time		Housework Time		Leisure Time	
	(1)	(2)	(3)	(4)	(5)	(6)
FMEPI	-1.126*** (0.247)	-0.553** (0.250)	0.932*** (0.115)	0.946*** (0.117)	-0.831*** (0.136)	-0.820*** (0.138)
Gender	7.906*** (0.252)	7.855*** (0.251)	-8.185*** (0.115)	-8.191*** (0.115)	1.192*** (0.148)	1.189*** (0.148)
Age	-0.194*** (0.011)	-0.164*** (0.011)	0.180*** (0.005)	0.186*** (0.005)	-0.148*** (0.006)	-0.166*** (0.007)
Hukou	-4.623*** (0.371)	-4.640*** (0.373)	-1.197*** (0.159)	-1.116*** (0.159)	2.559*** (0.226)	2.225*** (0.227)
Education	0.059* (0.034)	0.015 (0.035)	-0.154*** (0.016)	-0.151*** (0.016)	0.331*** (0.020)	0.314*** (0.020)
Status	-0.524*** (0.131)	-0.579*** (0.131)	0.299*** (0.063)	0.299*** (0.063)	0.038 (0.076)	0.042 (0.076)
Familysize		0.049 (0.111)		0.079 (0.053)		-0.258*** (0.066)
F_income		0.777*** (0.110)		-0.081* (0.049)		0.395*** (0.071)
Proportion16		0.127 (0.150)		0.015 (0.067)		0.146 (0.095)
Proportion65		-9.672*** (0.627)		-1.198*** (0.309)		3.057*** (0.364)
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
County FE	Yes	Yes	Yes	Yes	Yes	Yes
R-squared	0.162	0.168	0.219	0.219	0.126	0.130
Observations	38657	38657	38657	38657	38657	38657

Note: All columns in the table present regression results with robust standard errors. Year FE refers to year fixed effect. County FE refers to county fixed effect. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

4.2 Instrumental variable

The OLS baseline regression results in Table 2 may face endogeneity issue interference. This subsection two-stage least squares estimation(2SLS) is used to further explore the causal relationship between FMEPI and individual time allocation. In section 3, it was mentioned that the regression results may be affected by reverse causality. To rule out this endogeneity issue, we use the AEPS and elevation drop as instrumental variables. The indicator of elevation drop comes from the question “How many meters is the elevation drop between the highest point and the lowest point of the village?” in the 2014 questionnaire. Compared with the existing literature(Zhang et al.,2019; Nie et al.,2021), our instrumental variables are more likely to be reasonable and effective instrumental variables,especially for elevation drop index.The main logical for utilizing the

elevation drop as instrumental variable come from two reasons. First of all, it has better correlation with FMEPI, the greater the elevation drop will increase the difficulty of the construction of natural gas pipelines and power facilities, especially in western China. Therefore, we expect that the greater of the elevation drop, the more obvious the phenomenon of FMEPI. And our first stage results in Table 3 well support our expectation. The second is better satisfy the exogeneity and exclusivity. The geographical characteristic variables are the better exogenous variables, so the exogenous hypothesis is not a threat to our analysis. In terms of exclusivity, although the elevation of western China is generally higher, people live and work mostly in flat areas, such as small areas of basins and plains. Therefore, elevation drop between where residents live and work is small and random. In addition, with the development of Chinese convenient transportation, which made the elevation drop is no longer an issue for individual time allocation. From the above points of view, the elevation drop satisfies exclusivity.

Table 3 shows the results of the two-stage regression. Columns (1), (3) and (5) show the results of the first-stage regression, and it can be seen that the instrumental variables show a significant positive correlation with FMEPI. Meanwhile, the F-value in the first stage are all significantly greater than 10, indicating that there is no problem of weak instrumental variables. Therefore, the instrumental variables in this paper is appropriate. Columns (2), (4) and (6) present the results of the second-stage regression, indicating a negative relationship between FMEPI and working time with a significance level of 1%, a positive relationship between FMEPI and housework time with a significance level of 1%, a negative relationship between FMEPI and leisure time with a significance level of 1%.

Table 3

Regression results with AEPS and Elevation_drop as instrumental variables (2SLS regression).

Variables	Working Time		Housework Time		Leisure Time	
	(1)	(2)	(3)	(4)	(5)	(6)
FMEPI		-2.830*** (0.919)		3.016*** (0.429)		-5.410*** (0.519)
AEPS	0.923*** (0.020)		0.923*** (0.020)		0.923*** (0.020)	
Elevation_drop	0.005** (0.002)		0.005** (0.002)		0.005** (0.002)	
Individual char.	Yes	Yes	Yes	Yes	Yes	Yes
Family char.	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
County FE	Yes	Yes	Yes	Yes	Yes	Yes
Wald chi2 statistics	1236.46		1236.46		1236.46	
F statistics of first stage	1256.11		1256.11		1256.11	
R ²		0.041		0.157		0.046
Observations	20380	20380	20380	20380	20380	20380

Note: Wald chi2 statistics refers to Cragg-Donald Wald F statistic. F statistics of first stage refers to Kleibergen-Paap Wald F statistic. The Individual includes gender, age, education, hukou, status. The Family includes family size, F_income, proportion16, proportion65. Hansen J P_values of columns (4) and (6) are greater than 0.1, no over-identification problem exists. Hansen J P_value of columns (2) is less than 0.1, the results of our analysis using only EP_county as IV are also consistent. All columns in the table present regression results with robust standard errors.

Year FE refers to year fixed effect. County FE refers to county fixed effect. * $p<0.1$, ** $p<0.05$, *** $p<0.01$.

4.3 Robustness check

In this subsection, we perform three sets of robustness tests, including alternative measure of energy accessibility poverty, alternative threshold of energy affordability poverty, and alternative estimation model.

4.3.1 Alternative measure of energy accessibility poverty

Developing countries generally use accessibility to capture household energy poverty (Dogan et al., 2021). Fig 1 presents 4921, 4387, 3675 households, approximately accounting 46.10%, 40.30%, 33.55% that still using firewood or coal for cooking respectively across three years in the sample. Therefore, it is reasonable to utilize fuel type to identify the household energy poverty, the alternative household energy poverty measure is cooking fuel type (EP). A household is defined as energy poverty if it uses firewood or coal for cooking, EP = 1, otherwise 0. Table 4 shows a positive relationship between EP and housework time with a significance level of 1% and a negative relationship between EP and leisure time with a significance level of 1%, which is consistent with before. What noteworthy is that the relationship between EP and working time is not significant in Table 4. Thus, we conclude that the affordability in energy poverty is also an important factor need to be considered.

Table 4

Cooking Fuel Types and Individual Working Time, Housework Time, and Leisure Time (OLS regression).

Variables	Working Time	Housework Time	Leisure Time
	(1)	(2)	(3)
EP	0.314 (0.345)	1.433*** (0.158)	-1.429*** (0.190)
Individual	Yes	Yes	Yes
Family	Yes	Yes	Yes
Year FE	Yes	Yes	Yes
County FE	Yes	Yes	Yes
R ²	0.168	0.220	0.131
Observations	38571	38571	38571

Note: The Individual and Family char. are consistent with Table 3. All columns in the table present regression results with robust standard errors. Year FE refers to year fixed effect. County FE refers to county fixed effect. * $p<0.1$, ** $p<0.05$, *** $p<0.01$.

4.3.2 Alternative threshold of energy affordability poverty

The famous 10% threshold proposed by Boardman in 1991 is more suitable for developed countries (Zhang et.al, 2019). A proportion of households spend more than 15% or even 20% of their household income on energy in China (Fig 5). As a result, the 15% or 20% threshold maybe more powerful to capture the energy affordability poverty in China. Column (1), (3) and (5) in table5 report the regression results with 15% threshold, column (2), (4) and (6) show the regression results with 20% threshold, all the indicators are similar with the baseline results. This suggests that the effect of household energy poverty on individual time allocation is significant and that household energy poverty hinders sustainable development by increasing the amount of time spent on unpaid housework and decreasing the amount of time spent on work that creates social value and on leisure time that creates social welfare.

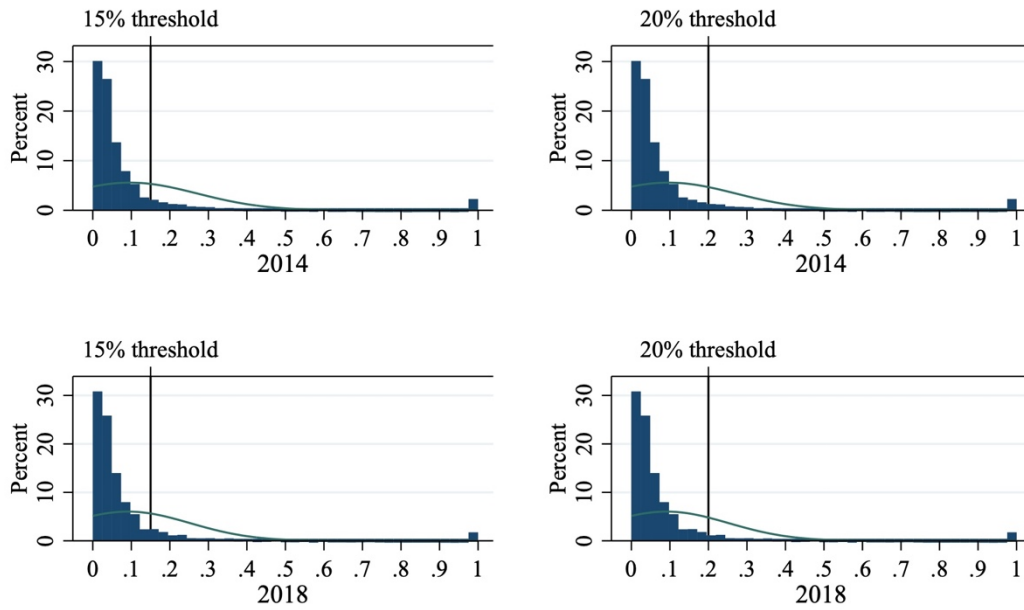


Fig. 5. The ratio of energy expenditure to household income (2014, 2018).

Source: China Family Panel Studies.

Table 5

Energy Poverty Level and Individual Working Time, Housework Time, and Leisure Time (OLS regression).

Variables	Working Time		Housework Time		Leisure Time	
	15%	20%	15%	20%	15%	20%
	(1)	(2)	(3)	(4)	(5)	(6)
Energy	-0.538**	-0.555**	0.937***	0.930***	-0.768***	-0.831***
Poverty Line	(0.269)	(0.282)	(0.126)	(0.133)	(0.149)	(0.156)
Individual	Yes	Yes	Yes	Yes	Yes	Yes
Family	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
County FE	Yes	Yes	Yes	Yes	Yes	Yes
R ²	0.168	0.168	0.219	0.219	0.130	0.130
Observations	38657	38657	38657	38657	38657	38657

Note: The Energy Poverty Level includes 15% and 20%. The Individual and Family char. are consistent with Table 3 .. All columns in the table present regression results with robust standard errors. Year FE refers to year fixed effect. County FE refers to county fixed effect. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

4.3.3 Alternative regression model

Tobit model was first proposed by economist James Tobin in 1958, and is suitable for analyzing data with significant truncation characteristics (Tobin, 1958). Tobit model takes account the concentration of observations at the limiting value when exploring the relationship between limited variable and other variables. Because the working time, housework time, and leisure time are all limited dependent variables, with a lower limit of 0. Therefore, it is reasonable to use the Tobit model for robustness test. The Tobit regression results reported in Table 6 show that the baseline results are robust to using this alternative model. Table 6 indicates a negative relationship between FMEPI and working time with a significance level of 5%, a positive relationship between FMEPI and housework time with a significance level of 1%, a negative relationship between FMEPI and

leisure time with a significance level of 1%. These findings are consistent with benchmark regression results.

Table 6

Energy Poverty and Individual Working Time, Housework Time, and Leisure Time (Tobit regression).

Variables	Working Time	Housework Time	Leisure Time
	(1)	(2)	(3)
FMEPI	-0.524** (0.285)	1.403*** (0.123)	-1.536*** (0.139)
Individual	Yes	Yes	Yes
Family	Yes	Yes	Yes
Year FE	Yes	Yes	Yes
County FE	Yes	Yes	Yes
χ^2	6989.49	8856.54	4791.42
Observations	38673	38673	38673

Note: The Individual and Family char. are consistent with Table 3 . All columns in the table present regression results with robust standard errors. Year FE refers to year fixed effect. County FE refers to county fixed effect. * $p<0.1$, ** $p<0.05$, *** $p<0.01$.

5. Heterogeneity analysis

This section will empirically investigate the heterogeneity in how household energy poverty affects individual time allocation, considering three key factors: gender, age, and Hukou type. To identify the statistical differences between all different groups (namely gender, age and Hukou), we also add interaction term analysis for both coefficient significant cases, as shown in Appendix.

5.1 Gender heterogeneity

In general, women may take on more housework and less work than men (Chen and Li, 2014; Blau and Kahn, 2017; Gao and Liu, 2023), but their leisure time is uncertain. Thus, the impact of household energy poverty on individual time allocation may vary by gender, we conduct group regressions based on gender (female=0, male=1) according to equation (1). Figure 6 displays the coefficients of gender heterogeneity analysis. Specifically, Panel A details that FMEPI significantly reduces men's working time but no substantial change in women's group. Panel B indicates that FMEPI will lead statistically significant increase in housework time for both the male group and female group. Interestingly, we observe that FMEPI increases more housework time of men compared with women. Then, we further investigate whether males increased significantly more time to housework compared to females by adding interaction term in Appendix Table 1, Column (5). We found that there is a statistically significant difference in housework increase between men and women. Thus, we could conclude that FMEPI narrows the gender gap in time spent on housework. Panel C shows that FMEPI reduces leisure time significantly for both the female group and the male group. Appendix Table 1, Column (8) checks whether there exists sufficient difference across gender groups, but does not find a significant difference. Across all Panels, we can observe the impact of FMEPI on individual working time and housework time performs statistically different, but the effects on leisure time appear roughly equivalent across gender groups. The decrease in working time and increase in housework time for men narrow the traditional gender gap, but it does not indicate that FMEPI promotes gender equality. Instead, other reasons are behind this phenomenon. In the following mechanism section, our findings support the fact that household energy poverty more likely diminishes the full-time work opportunities of men, resulting in a waste

of men's human capital. Furthermore, household energy poverty also tends to elevate the rate of singleness among men, thus leading to an increase in men's time devoted to housework.

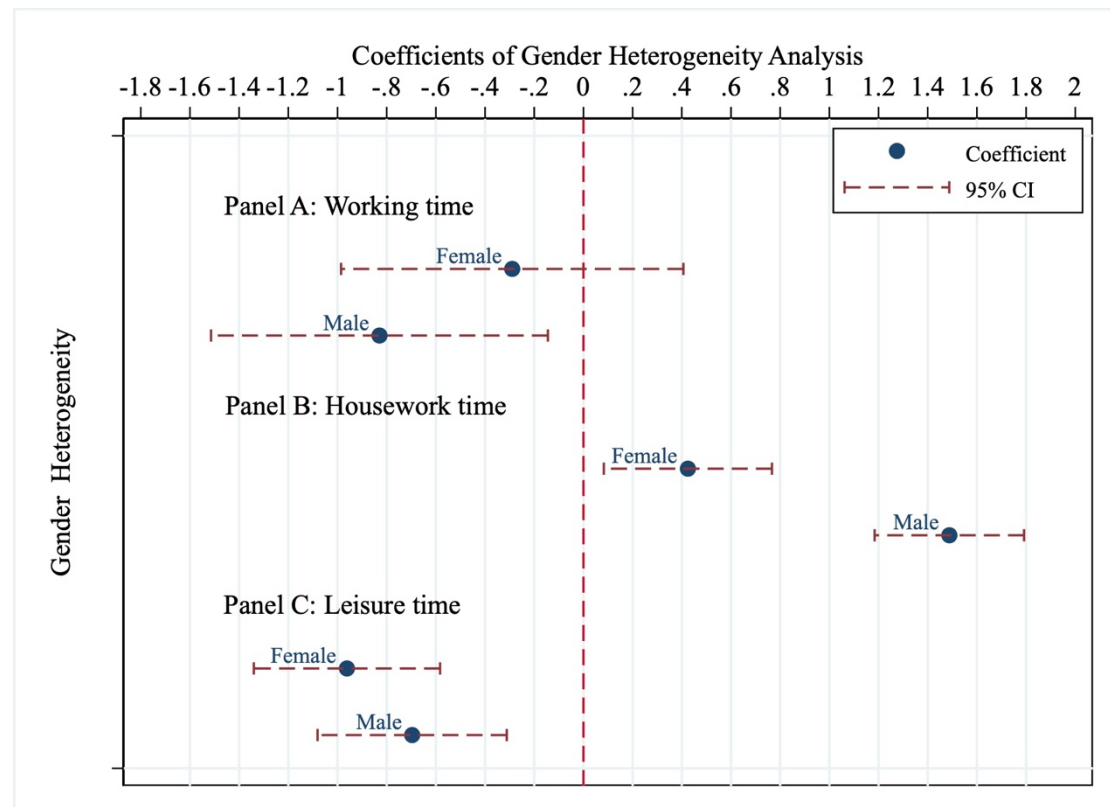


Fig. 6. The coefficients of gender heterogeneity analysis.

5.2 Age heterogeneity

According to the general boundary 45-year-old, this paper divides the sample into a younger group and an older group and analyses the heterogeneity across age groups. Specifically, the age group less than 45 years old is classified as the younger group (defined $age_goup=1$), and the age group greater than or equal to 45 years old is classified as the older group ($age_goup=0$). Figure 7 summarizes the estimation results of age heterogeneity analysis, and all regression settings remain consistent with before. Panel A assesses the influence of FMEPI on personal working time across age groups. Here, we also find an interesting phenomenon that FMEPI reducing working time mainly occurs among the younger group. The youth group is the dominant group in social development, but FMEPI reduces their working time, which is an important social issue that deserves attention. Panel B shows the significant impact of FMEPI on individual housework time for both younger and older groups. Appendix Table 2, Column (5) checks the statistical difference across age groups, then the estimation results indicate that FMEPI significantly reduces more housework for people who are below 45 years old. The possible reason comes from the different family sizes, in other words, the average family size of the older group is greater than the younger group. The data shows that the average family size for individuals over 45 years of age is 4.7 compared to 4.0 for those under 45 years of age, which supports this phenomenon. Panel C indicates a significant decrease in leisure time for both younger and older groups. The above findings indicate that focusing on and improving the issue of FMEPI is of great relevance to the youth people and is of essential to social development.

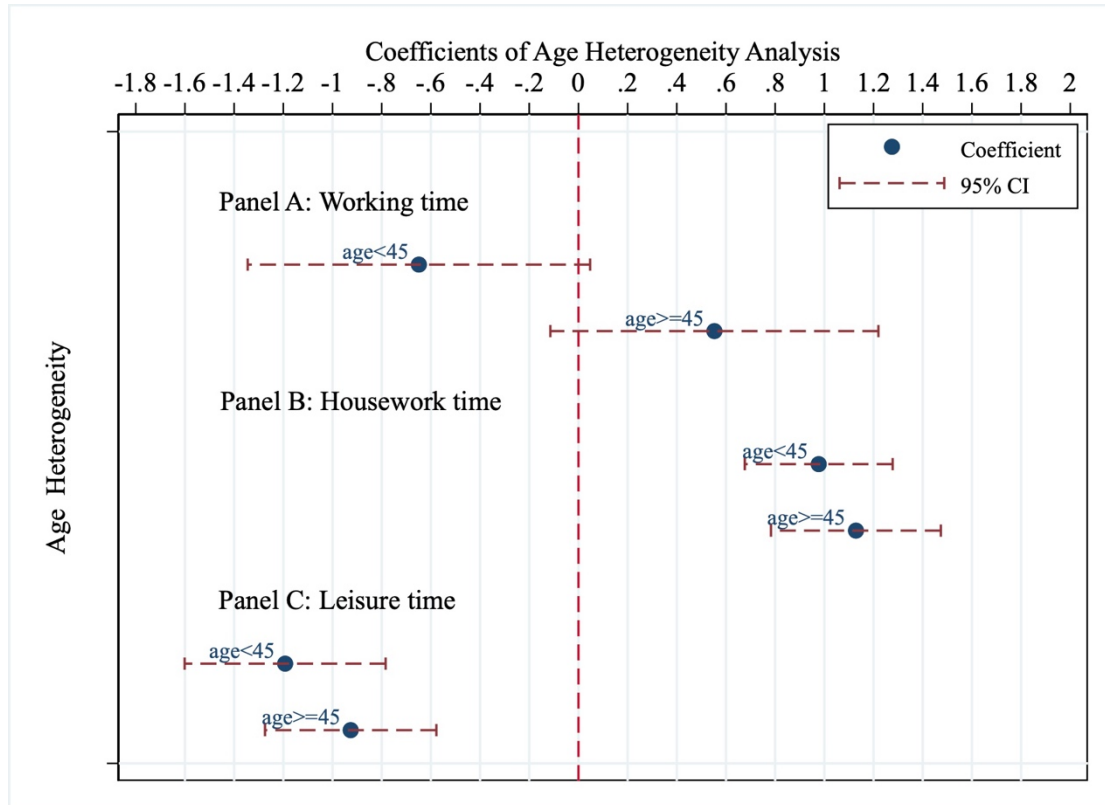


Fig. 7. The coefficients of age heterogeneity analysis.

5.3 Hukou heterogeneity

China's unique hukou system allows non-agricultural hukou groups to have better access to better basic public services and employment opportunities, which may make the impact of household energy poverty on the distribution of time vary by Hukou type. Therefore, this study conducts group regressions based on agricultural and non-agricultural Hukou household heterogeneity, we defined that non-agricultural Hukou equals 1. Figure 8 shows the estimated results of analyzing the impact of FMEPI on individual time allocation with different Hukou.

Panel A shows a negative relationship between FMEPI and working time for agricultural Hukou individuals. However, there is no significant effect on the non-agricultural population, which suggests that the loss of working time due to FMEPI mainly comes from agricultural Hukou families. The potential reason may come from where they live. In China, people living in rural areas have agricultural Hukou, and people living in urban areas have non-agricultural Hukou, and a person cannot have both types of Hukou. As a result, people living in urban areas with non-agricultural Hukou have more job opportunities than those who have agricultural Hukou, so energy poverty does not have a significant impact on the working time of people with non-agricultural Hukou. Similarly, we consistently observe that FMEPI influences the housework time and leisure time of individuals in both agricultural and non-agricultural Hukou individuals significantly (Panel A and Panel B) and the effects are roughly equal (Appendix Table 3).

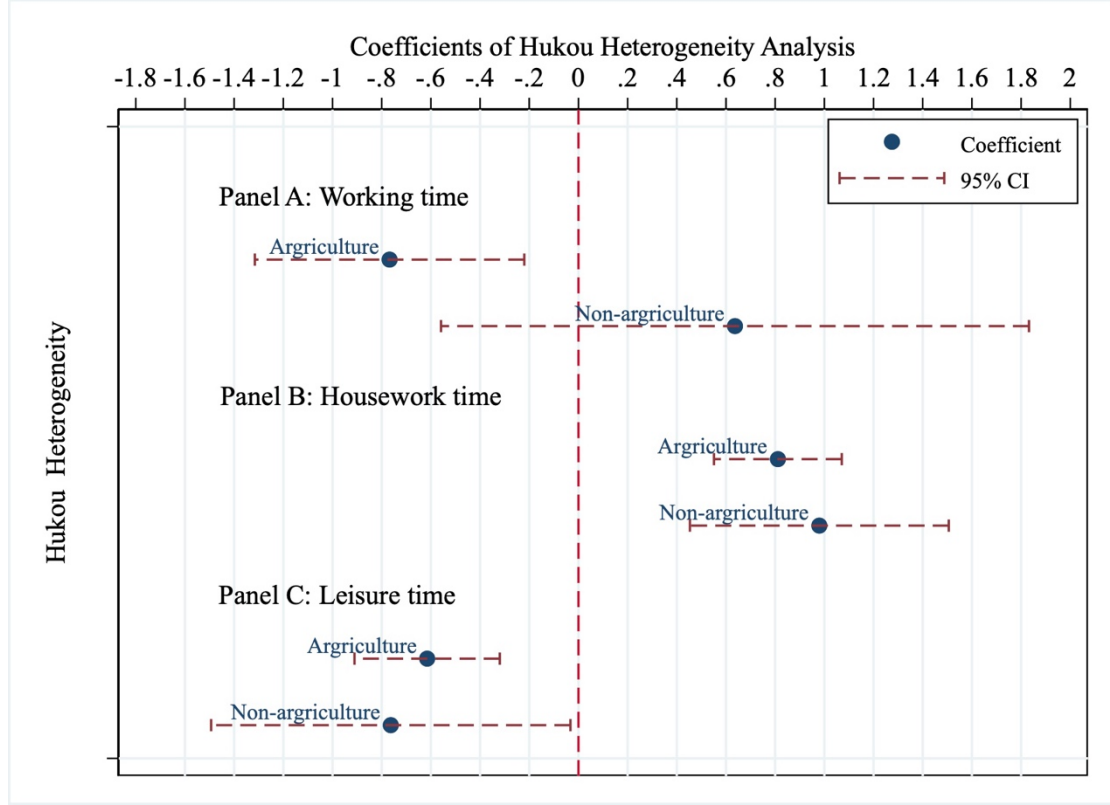


Fig. 8. The coefficients of Hukou heterogeneity analysis.

6. Mechanism verification

This paper has demonstrated that FMEPI and individual time allocation are indeed strongly correlated, but we are interested in the potential mechanisms. Given data availability, we are not able to explore all the mechanisms by which household energy poverty affects time allocation. Nevertheless, we attempted to explore the mechanisms in four areas: individual attitudes to life, individual health, individual employment, and marital status. Since the mediation effect model based on the stepwise method is not reliable in economics for mediation effect testing, there are problems such as endogeneity bias and unclear identification of some channels (Jiang, 2022). Therefore, this study tests the effect of mechanism variables by observing the core independent variables referring to Jiang (2022). The mechanism effect model is specifically set as:

$$IM = \alpha + \beta FMEPI_{cit} + \gamma X_{cit} + \mu_i + \tau_t + \varepsilon_{cit} \quad (2)$$

where IM is the mechanism variable, including life satisfaction, future confidence, health status, employment type, and single status (Fig. 9). Other terms are consistent with equation (1). Given that we mainly focus on the existence of mechanisms in different groups, but do not care about the heterogeneity differences of mechanisms. Therefore, in the mechanism analysis, we will not do interaction term analysis as in the heterogeneity analysis section.

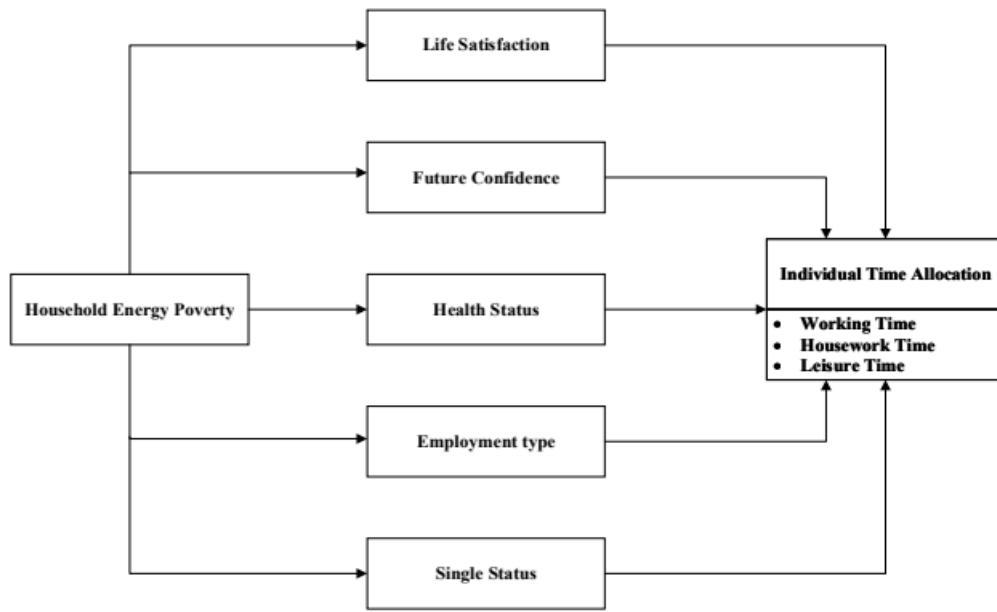


Fig. 9. The framework of the research mechanism.

Firstly, household energy poverty may bring about negative attitudes towards life, which may affect an individual's motivation to work and live, thereby reducing the allocation of personal working and leisure time. Some scholars have found that household energy poverty affects an individual's subjective well-being directly. For instance, children and adults in households suffering from household energy poverty have a greater probability of depression and unhappiness (Zhang et al., 2021; Nie et al., 2021). A healthy psychological condition is a requirement to guide individuals to work and enjoy life positively, so negative attitudes towards life resulting from household energy poverty are likely to be an important channel influencing the allocation of time. According to the CFPS adult self-response database "How satisfied are you with your life in your own home?" and "How confident are you about your future?" These two questions capture indicators of life satisfaction and future confidence at the subjective level of the individual household. The minimum value of the life satisfaction indicator is 1, which means very dissatisfied with life, and the maximum value is 5, which means very satisfied with life. The minimum value of the future confidence indicator is 1, which represents very little confidence in the future, and the maximum value is 5, which represents very much confidence in the future.

Table 7 shows the relationship between FMEPI and life satisfaction and future confidence based on three panels, including gender, age, and urban-rural group. Columns (1)-(3) of Panel A report the statistically significant negative relationship between FMEPI and life satisfaction based on gender group. Columns (4)-(6) of Panel A reveal the negative relationship between FMEPI and future confidence. The reduction in life satisfaction may reduce motivation to participate in leisure activities, further leading to the decrease in leisure time, because of the significant relationship between leisure engagement and subjective well-being (Kuykendall and Ng, 2015). Additionally, the decrease in future confidence may be the path to the reduction in working time, as the positive relationship between future confidence and working time (Judge and Bono, 2001).

Columns (1)-(3) of Panel B show the significantly negative relationship between FMEPI and life satisfaction based on age group. Columns (4)-(6) of Panel B indicate coefficient estimate close

to zero and statistically significant, except in column (5), suggesting the overall sample significant negative correlation may come from the group greater than or equal to 45 years of age.

Columns (1)-(3) of Panel C show the significantly negative relationship between FMEPI and life satisfaction based on urban-rural group. Columns (4)-(6) of Panel C reveal the negative relationship between FMEPI and future confidence based on urban-rural group, especially the individual who lived in urban suffers a worse reduction in future confidence, further result in a decreasing in working time.

Table 7

Mechanism analysis (Life Satisfaction and Future Confidence).

Variables	Life Satisfaction			Future Confidence		
	(1)	(2)	(3)	(4)	(5)	(6)
Panel A: Gender grouping regression						
	All	Female	Male	All	Female	Male
FMEPI	-0.066*** (0.010)	-0.061*** (0.014)	-0.070*** (0.014)	-0.055*** (0.010)	-0.059*** (0.014)	-0.051*** (0.013)
Individual	Yes	Yes	Yes	Yes	Yes	Yes
Family	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
County FE	Yes	Yes	Yes	Yes	Yes	Yes
R ²	0.149	0.135	0.170	0.124	0.119	0.137
Observations	38657	18903	19754	38657	18903	19754
Panel B: Age grouping regression						
	All	Age<45	45<=Age	All	Age<45	45>=Age
FMEPI	-0.066*** (0.010)	-0.062*** (0.014)	-0.082*** (0.014)	-0.055*** (0.010)	-0.025** (0.013)	-0.070*** (0.015)
Individual	Yes	Yes	Yes	Yes	Yes	Yes
Family	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
County FE	Yes	Yes	Yes	Yes	Yes	Yes
R ²	0.149	0.149	0.165	0.124	0.099	0.141
Observations	38657	19784	18873	38657	19784	18873
Panel C: Hukou grouping regression						
	All	Agriculture	Non-agr.	All	Agriculture	Non-agr.
FMEPI	-0.066*** (0.010)	-0.049*** (0.013)	-0.078*** (0.016)	-0.055*** (0.010)	-0.032** (0.013)	-0.075*** (0.016)
Individual	Yes	Yes	Yes	Yes	Yes	Yes
Family	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
County FE	Yes	Yes	Yes	Yes	Yes	Yes
R ²	0.149	0.159	0.148	0.124	0.141	0.117
Observations	38657	19441	18782	38657	19441	18782

Note: The Individual and Family char. are consistent with Table 3.. All columns in the table present regression results with robust standard errors. Year FE refers to year fixed effect. County FE refers to county fixed effect. * $p<0.1$, ** $p<0.05$, *** $p<0.01$.

Secondly, household energy poverty may affect the status of health of individuals, leading to

changes in time allocation. A healthy body is a prerequisite for normal work and a necessary guarantee for active enjoyment of life. Notably, Rafi et al. (2021) have explored the statistically significant negative impact of energy poverty on individual health. However, household energy poverty not only affects mental health but also has a significant impact on an individual's physical health (Zhang et al., 2021). Households suffering from energy accessibility widely use traditional stoves (without a chimney or grate) to cook, which is the leading cause of indoor air pollution (Tang and Liao, 2014). Although the Ministry of Agriculture (MOA) launched the National Improved Stove Program (NISP) to deliver clean and efficient stoves to rural households, more than 50% of rural households do not receive this improved stove, or those who do have problems with an inappropriate way (Sinton et al., 2004; Tang and Liao, 2014; Hanna et al., 2016;). In the iteration of the Global Burden of Disease (GBD) project (Lim et al., 2012), it was projected that roughly 1.04 million individuals in China experience premature mortality due to indirect consequences arising from indoor air pollution caused by the combustion of solid fuels. Additionally, the indoor air pollution caused by the stove exerts a significant negative impact on well-being ((Duo et al., 2008). Furthermore, the physical health problems caused by indoor air pollution may make people less efficient in housework, the resulting psychological problems may make people less enthusiastic about leisure activities, and these health problems may also reduce the opportunities of obtaining stable employment. Therefore, health may be one mechanism of how household energy poverty affects individual time allocation.

An individual's physical health status can be obtained from the question "Respondent's health status" in the CFPS database. The minimum value of the physical health indicator is 1 which means very bad, and the maximum value is 7 which means very good. Based on this question, the paper went on to explore the impact of household energy poverty on the health status of individuals, and the results are shown in Table 8. Columns (1)-(3) of Panel A report the significantly negative relationship between FMEPI and health, which is consistent with Zhang et al. (2019), Churchill and Smyth (2021).

Columns (1)-(3) of Panel B display the estimated coefficients of the impact of household energy poverty on health status for the alternative age group. The figures consistently show a statistically significant positive relationship, and the magnitude of the coefficient estimate becomes smaller as the age increases. This aligns with common sense that the influence of household energy poverty is accumulative over time, thus individuals who aged above or equal to 45 experience a worse reduction in health.

Columns (1)-(3) of Panel C examine the relationship between FMEPI and health based on urban-rural groups. Just look at the regression estimators, the individual who live in urban suffers a worse reduction in health, the possible reason is that urban households are more reliable on modern energy to maintain suitable temperatures in extreme temperature weather, but households that experience energy poverty may not be sufficient to meet these needs, which may result in a worse reduction in health for urban households.

Table 8

Mechanism analysis (Health Status)

Variables	Health Status	Health Status	Health Status
	(1)	(2)	(3)
Panel A: Gender grouping regression			
	All	Female	Male
FMEPI	-0.073*** (0.010)	-0.080*** (0.015)	-0.070*** (0.015)
Individual	Yes	Yes	Yes
Family	Yes	Yes	Yes
Year FE	Yes	Yes	Yes
County FE	Yes	Yes	Yes
R ²	0.236	0.267	0.212
Observations	38760	18971	19789
Panel B: Age grouping regression			
	All	Age<45	45>=Age
FMEPI	-0.073*** (0.010)	-0.048*** (0.014)	-0.084*** (0.016)
Individual	Yes	Yes	Yes
Family	Yes	Yes	Yes
Year FE	Yes	Yes	Yes
County FE	Yes	Yes	Yes
R ²	0.236	0.203	0.211
Observations	38760	19818	18942
Panel C: Hukou grouping regression			
	All	Agriculture	Non-agriculture
FMEPI	-0.073*** (0.010)	-0.049*** (0.014)	-0.070*** (0.017)
Individual	Yes	Yes	Yes
Family	Yes	Yes	Yes
Year FE	Yes	Yes	Yes
County FE	Yes	Yes	Yes
R ²	0.236	0.264	0.229
Observations	38760	19501	18825

Note: The Individual and Family char. are consistent with Table 3.. All columns in the table present regression results with robust standard errors. Year FE refers to year fixed effect. County FE refers to county fixed effect. * $p<0.1$, ** $p<0.05$, *** $p<0.01$.

Furthermore, energy poverty has the potential influence on individuals' employment types, consequently impacting how they allocate their time. As demonstrated in the earlier analyses, energy poverty can notably compromise an individual's mental and physical health, precipitating various alterations in their personal and family life dynamics. Diminished life satisfaction and worse physical health tend to discourage individuals from active participation in the workforce. While acknowledging the possibility of reciprocal causation among these factors, this study contends that this pathway of influence remains plausible. Moreover, energy poverty may indirectly contribute to reduced employment participation by affecting an individual's cognitive capabilities, among other

factors. Consequently, this study aims to clarify whether the curtailed employment opportunities are indeed the root cause of the decline in individuals' work time. Using a binary approach with a value of 1 assigned to individuals engaged in full-time employment and 0 to those who are employed in part-time occupation. Table 9 details the impact of energy poverty on individual employment type.

Columns (1)-(3) of Panel A in Table 9 show the significantly negative relationship between FMEPI and employment type based on gender group, suggesting that the impact of energy poverty on employment type is similar for both men and women. Columns (1)-(3) of Panel B display that the overall sample significant negative correlation comes from the group greater than or equal to 45 years of age. The possible reason may come from the health vulnerabilities of older people, this vulnerability is more likely to occur in the state of family energy poverty. Therefore, energy poverty negatively affects access to full-time employment for people aged above or equal to 45, but has no significant effect on those under 45. Columns (1)-(3) of Panel C estimate the relationship between FMEPI and employment type based on Hukou group. Results of Panel C display that the impact of energy poverty on employment are negative for both agricultural-Hukou people and non-agriculture-Hukou people, which suggests that the impact of household energy poverty on employment is consistent across Hukou groups.

Table 9

Mechanism analysis (Employment Type)

Variables	Employment Type (1)	Employment Type (2)	Employment Type (3)
Panel A: Gender grouping regression			
	All	Female	Male
FMEPI	-0.195*** (0.035)	-0.177*** (0.056)	-0.211*** (0.045)
Individual	Yes	Yes	Yes
Family	Yes	Yes	Yes
Year FE	Yes	Yes	Yes
County FE	Yes	Yes	Yes
Observations	38657	18723	19754
Panel B: Age grouping regression			
	All	Age<45	45>=Age
FMEPI	-0.195*** (0.035)	-0.033 (0.042)	-0.411*** (0.066)
Individual	Yes	Yes	Yes
Family	Yes	Yes	Yes
Year FE	Yes	Yes	Yes
County FE	Yes	Yes	Yes
Observations	38657	19784	18734
Panel C: Hukou grouping regression			
	All	Agriculture	Non-agriculture
FMEPI	-0.195*** (0.035)	-0.112** (0.054)	-0.146*** (0.050)
Individual	Yes	Yes	Yes
Family	Yes	Yes	Yes
Year FE	Yes	Yes	Yes

County FE	Yes	Yes	Yes
Observations	38657	19432	18683

Note: The Individual and Family char. are consistent with Table 3.. All columns in the table present regression results with robust standard error. Year FE refers to year fixed effect. County FE refers to county fixed effect. * $p<0.1$, ** $p<0.05$, *** $p<0.01$.

Finally, individual time allocation is possibly related to their single status, and household energy poverty can alter this allocation by impacting the single status of a person. In the case of a single individual, they may need to allocate more of their time to housework. However, when an individual is married, the presence of a spouse can facilitate the division of household tasks, thereby introducing household size effects and reducing time inefficiencies. Therefore, this study delves into the deeper connection between household energy poverty and single status. Generally, regions suffering household energy poverty are characterized by both underdeveloped economies and unsanitary household living conditions, both of which can significantly influence the ease with which adults can enter into marriage. Consequently, this research conducts an analysis based on individual marital status, classifying the unmarried and widowed individuals as the single group, and the married and remarriage individuals as the non-single group then regresses according to gender group.

Columns (1)-(3) of Panel A in Table 10 conduct the regression analysis based on gender group, which suggests that energy poverty households have higher single rates for both men and women. Panel B indicates the significantly positive relationship between FMEPI and single status based on age group, except Column (3), suggesting the overall sample significant positive correlation comes from the group below 45 years of age. This aligns with the social context in China. As China's economy grows and people's living conditions generally improve, females usually require males to have a house, car, and deposits when they marry⁵. This has significantly increased the rate of single among younger males under the age of 45 who suffer household energy poverty. Panel C reports the significantly positive relationship between FMEPI and single status based on the urban-rural group. The estimated coefficient of people who live in rural areas is slightly bigger than in urban, the potential reason comes from the relatively higher opportunities to marry and avoid divorcing than in rural households may be due to the higher income and cleaner living conditions, which are tied to family energy poverty status.

⁵ 71.8% of Chinese women argue that a man is eligible to marry only if he owns a house in 2020. Source: <https://zhuanlan.zhihu.com/p/142357411>

Table 10

Mechanism analysis (Single Status)

Variables	Single Status	Single Status	Single Status
	(1)	(2)	(3)
Panel A: Gender grouping regression			
	All	Female	Male
FMEPI	0.199*** (0.028)	0.197*** (0.040)	0.222*** (0.042)
Individual	Yes	Yes	Yes
Family	Yes	Yes	Yes
Year FE	Yes	Yes	Yes
County FE	Yes	Yes	Yes
Observations	38657	18903	19754
Panel B: Age grouping regression			
	All	Age<45	45<=Age
FMEPI	0.199*** (0.028)	0.246*** (0.044)	-0.053 (0.052)
Individual	Yes	Yes	Yes
Family	Yes	Yes	Yes
Year FE	Yes	Yes	Yes
County FE	Yes	Yes	Yes
Observations	38657	19784	18859
Panel C: Hukou grouping regression			
	All	Agriculture	Non-agriculture
FMEPI	0.199*** (0.028)	0.208*** (0.038)	0.143*** (0.047)
Individual	Yes	Yes	Yes
Family	Yes	Yes	Yes
Year FE	Yes	Yes	Yes
County FE	Yes	Yes	Yes
Observations	38657	19429	18764

Note: The Individual and Family char. are consistent with Table 3.. All columns in the table present regression results with robust standard errors. Year FE refers to year fixed effect. County FE refers to county fixed effect. * $p<0.1$, ** $p<0.05$, *** $p<0.01$.

7. Conclusions and implications

Household energy poverty not only has an impact on the economic development of society, but also changes the way people work and live, thereby affecting individual time allocation, which is of profound significance to the sustainable development of the society. Accordingly, this study examines the effect of household energy poverty on individual time allocation based on the data of CFPS. Secondly, this paper also analyses the heterogeneity of the effects of household energy poverty. Finally, the study also explores the potential impact mechanisms. The main contribution of this study is that it explores the relationship between household energy poverty and individual time allocation for the first time, and these findings provide a reference for eliminating the waste of human resources, increasing individual well-being and improving economic development of society.

The main findings of this study are listed as follows. Firstly, the household energy poverty significantly reduces individual working time and leisure time and increases time spent on housework, resulting in a waste of human resources. Secondly, the heterogeneity analysis finds that the influence of household energy poverty on individual time allocation differs across gender, age, and urban-rural areas. Household energy poverty decreases more working time of men, reduces more working time of younger people, and likely leads to possible reductions in working time for people who has agricultural Hukou, resulting in more wasted human resources for economic development. Interestingly, for men and younger people, household energy poverty can increase more housework time and decrease more leisure time. Finally, this paper also identifies individual life attitudes, health level, employment status, and single status as four potential mechanisms by which household energy poverty affects time allocation, and these mechanisms also provide additional evidence of the significant impacts of household energy poverty. We found that household energy poverty has a negative impact on life attitudes and health, further negatively affects employment. It also increases the possibility of being single, thus influencing individual housework workload.

A heightened awareness of the significance of addressing household energy poverty on individual behavior and social development is a crucial part of sustainable development. The newly founds effect of household energy poverty on individual time allocation and its relevant mechanisms buttresses a perspective for ameliorating a waste of human resources and improving the economic development of society. However, the empirical evidence comes from the largest developing country, where the well-being of effective time allocation is homogeneous for human beings worldwide. We believe that the findings can also provide a reference for other developing countries. Therefore, our results have specific policy implications for similar developing countries that suffer from household energy poverty.

Firstly, it is necessary to improve the accessibility to modern energy. Although the coverage of electricity has reached 100% in China, insufficient and unstable electricity supply problems exist, especially in remote areas. District with these electricity problems may lack attraction for factories to locate, which reduces employment opportunities and wastes human resources in the area. In terms of cooking fuels in households, there is a noticeable proportion of households use firewood or coal to cook, which increases housework time and crowd out leisure time. Thus, the reduction in leisure time may lower individual well-being. Therefore, we advocate that governments can enhance energy infrastructure to abjure the accessible problem.

Secondly, governments are supposed to pay more attention to energy poverty among male groups according to the heterogeneity analysis for gender. The results of gender heterogeneity analysis indicate that household energy poverty significantly decreases the working time of males, which catalyzes the more significant increase in housework time for males. In addition, the people who suffer from household energy poverty lack attraction for the opposite sex, so they do not have a spouse to share housework, which leads to a greater increase in housework time. Furthermore, the rise in male housework time may lead to a waste of high-quality labor, which is also detrimental to economic development.

Thirdly, the issue of energy poverty among the younger group also demands urgent attention. The results of age heterogeneity analysis imply that people younger than 45 years old experience a worse working time reduction due to household energy poverty. The government could consider subsidizing energy consumption for energy-poor households to encourage the use of clean energy

by young people. In addition, it is necessary to properly guide them to find suitable employment and fundamentally solve the problem of energy poverty. With the promotion of the digital economy, the authorities can provide subsidies to promote Internet applications, further enhance their digital skills and office skills, and facilitate the delivery of recruitment information to promote stable employment.

In conclusion, this study examines the relationship between household energy poverty and individual time allocation, conducts several heterogeneity analyses, and identifies four potential mechanisms to provide a reference to understand how household energy poverty affects individual time allocation.

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